

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. We extract the exact particle-number coefficient before Fourier inversion of continuous constraints, thereby removing the grand-canonical empty-sector atom. A coefficient-level trajectory-polymer theorem proves that either linearly many bounded regular components are present or the coefficient weight is exponentially small. This yields a compact-annulus gap and an integrable high-frequency bound whose power grows linearly with particle number, without any impossible spatial block independence. Exact activity and continuous saddles, a quantitatively defined regular shell class, the mixed local coefficient, and the source-dependent microcanonical pressure and covariance Schur complement follow.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: EXACT-NUMBER POLYMER COEFFICIENTS AND REGULAR SHELLS

The grand-canonical zero-particle atom is never Fourier inverted in the continuous constraint variables. Particle number is extracted first by a Cauchy coefficient at its exact finite-volume saddle. The remaining canonical characteristic function is analyzed by a labelled trajectory polymer expansion; no spatial independence at a growing propagation radius is assumed.

1.1. The coefficient-level polymer gas. Let z be the activity and let H denote the complete bounded particle-path and actual-contact source supplied by the grand-canonical part of B2. For a continuous constraint mark $Y \in \mathbb{R}^d$, put

$$\mathcal{Z}_\varepsilon(z, u; H) = \mathbb{E}_{\text{Pois}} \left[z^{N_\varepsilon} e^{\mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle + iu \cdot \sum_{j=1}^{N_\varepsilon} Y_j} 1_{\text{hc}} \right].$$

The connected trajectory expansion has the labelled exponential form

$$\mathcal{Z}_\varepsilon(z, u; H) = \sum_{N \geq 0} \frac{z^N}{N!} \sum_{\Pi \in \mathfrak{P}_N} \prod_{C \in \Pi} \zeta_{\varepsilon, H, u}(C),$$

where C is a connected labelled trajectory polymer. On the short-time source chart, B2 proves the Kotecký–Preiss estimate

$$\sup_i \sum_{C \ni i} e^{a|C|} |\zeta_{\varepsilon, H, 0}(C)| \leq a_0 < 1.$$

Lemma 1.1 (Many regular components or exponential penalty). *There are $m_0, c_0, c_1 > 0$, and a finite collection of regular polymer types $\mathcal{G}$, each of size at most m_0 , such that in the exact- N coefficient:*

$$\frac{1}{N!} [z^N] \mathcal{Z}_\varepsilon(z, 0; H)$$

the total absolute weight of partitions containing fewer than $c_0 N$ members of $\mathcal{G}$ is at most $e^{-c_1 N}$ times the total positive coefficient. On every $G \in \mathcal{G}$, the summed mark map has a smooth density whose characteristic function satisfies

$$|\hat{\nu}_G(u)| \leq C_s (1 + |u|)^{-s}$$

for every fixed s , uniformly in the source chart.

Proof. Mark every polymer in $\mathcal{G}$ by an auxiliary variable v . The coefficient pressure

$$p_N(v) = \frac{1}{N} \log [z^N] \mathcal{Z}_\varepsilon(z, 0; H; v)$$

is analytic for $|v - 1| < \delta$, by the same polymer norm. The singleton and finitely many two-particle regular collision polymers give $\partial_{\log v} p_N(1) \geq 2c_0$: their activities are uniformly positive on fixed position and velocity patches, while the total activity of polymers of size $> m_0$ is made smaller than c_0 by the exponential size tail. Chernoff's inequality at $v < 1$ gives the exponential lower-tail estimate.

For a regular singleton, the velocity/position mark map has full-rank Jacobian by the declared affine-span hypothesis. When energy and momentum make the singleton singular, use one of the finitely many regular $2, \dots, m_0$ -particle polymers; the convolution map is a submersion on a compact positive-activity patch. Coarea gives a C^∞ density there, and repeated integration by parts gives the Fourier estimate. The finite family covers every direction $u/|u|$. $\square$

This lemma is combinatorial and coefficient-level. It neither partitions the torus into spatially independent boxes nor discards high velocities.

1.2. Exact particle-number extraction. Let $N_\varepsilon/\mu_\varepsilon \rightarrow n_0 > 0$. Define

$$Z_{\varepsilon,N}(u; H) = [z^{N_\varepsilon}] \mathcal{Z}_\varepsilon(z, u; H).$$

Let $z_{\varepsilon,H,u}$ solve the finite-volume coefficient saddle

$$z \partial_z \log \mathcal{Z}_\varepsilon(z, u; H) = N_\varepsilon.$$

Theorem 1.2 (Uniform coefficient saddle). *On a common complex neighborhood of $u = 0$ there is a unique saddle $z_{\varepsilon,H,u}$, analytic in (H, u) , and*

$$\frac{1}{\mu_\varepsilon} \log Z_{\varepsilon,N}(u; H) = \frac{1}{\mu_\varepsilon} \log \mathcal{Z}_\varepsilon(z_{\varepsilon,H,u}, u; H) - \frac{N_\varepsilon}{\mu_\varepsilon} \log z_{\varepsilon,H,u} + o(1)$$

locally uniformly with the first three derivatives. The activity Hessian is bounded above and below uniformly.

Proof. Under the coefficient-positive real law, $z \partial_z \log \mathcal{Z}$ is the mean particle number and its derivative is the variance. Lemma 1.1 and the positive singleton activity give a variance between $c\mu_\varepsilon$ and $C\mu_\varepsilon$. The analytic implicit-function theorem gives the saddle. Cauchy's formula on $|z| = |z_{\varepsilon,H,u}|$ has a Gaussian central arc. Outside that arc, the span-one particle count and the polymer aperiodicity give an exponential gap. Steepest descent yields the coefficient formula and derivative convergence. $\square$

After this extraction the particle number is fixed and is not included as a Fourier coordinate. Let

$$\varphi_{\varepsilon,N,H}(u) = \frac{Z_{\varepsilon,N}(u; H)}{Z_{\varepsilon,N}(0; H)} e^{-iu \cdot \mathbb{E}_{N,H} \sum_j Y_j}.$$

Theorem 1.3 (Canonical full-frequency theorem). *There are $\delta < R < \infty$ and $c > 0$ such that:*

$$\begin{aligned} |u| \leq \delta : & \quad |\varphi_{\varepsilon,N,H}(u)| \leq e^{-cN|u|^2}, \\ \delta \leq |u| \leq R : & \quad |\varphi_{\varepsilon,N,H}(u)| \leq e^{-cN}, \\ |u| > R : & \quad |\varphi_{\varepsilon,N,H}(u)| \leq e^{-cN} + C_s(1 + |u|)^{-s-c_0N}. \end{aligned}$$

The estimates are uniform on compact source and target charts. In particular, for $s > d + 3$,

$$\int_{|u| > R} |\varphi_{\varepsilon,N,H}(u)| du = o(N^{-d/2}).$$

Proof. The central estimate is the coefficient Hessian bound from Theorem 1.2, including every cross term. On the compact annulus, equality in modulus would make $u \cdot Y$ constant modulo 2π on every regular polymer type. The affine-span assumption excludes this; compactness and coefficient pressure analyticity give an exponential gap.

For the tail, split coefficient partitions according to Lemma 1.1. The exceptional class has relative weight e^{-cN} . Every remaining partition contains c_0N disjoint regular components. Condition on all other components and integrate their marks; each contributes the smooth Fourier factor. Using one integration power per regular component and reserving s extra derivatives gives the displayed exponent. The exponential term is a coefficient weight, not a frequency-independent integrand: on the exceptional class its continuous mark law still has one fixed regular component, so it is multiplied by $C_s(1 + |u|)^{-s}$. Thus the actual bound is

$$C_s \left[e^{-cN} (1 + |u|)^{-s} + (1 + |u|)^{-s - c_0N} \right],$$

whose integral has the stated order. This also treats all singular low-particle grand-canonical sectors, because they are absent after exact N -extraction. $\square$

1.3. Continuous constraints and the source-dependent saddle. Let C denote momentum, energy, and the declared additional continuous constraints, all expressed per particle. For target a in the relative interior of their affine support, let $\lambda_{\varepsilon, H, a}$ solve

$$D_\lambda Q_{\varepsilon, N}(H, \lambda) = a, \quad Q_{\varepsilon, N} = \frac{1}{\mu_\varepsilon} \log Z_{\varepsilon, N}(0; H + C^* \lambda).$$

Theorem 1.4 (Finite source-dependent continuous saddle). *The map $(H, a) \mapsto \lambda_{\varepsilon, H, a}$ exists uniquely on a common regular chart and converges, with two derivatives, to $\lambda_{H, a}$. Its Hessian is the full canonical covariance matrix, including all cross terms, and*

$$cI \leq D_{\lambda\lambda}^2 Q_{\varepsilon, N} \leq cI.$$

Proof. The regular component family in Lemma 1.1 contains full-rank mark patches in every constraint direction. Since there are linearly many such components in the coefficient law, their conditional covariance is at least cN ; the polymer moment estimate gives the upper bound. Properness on the relative interior follows by sending a multiplier to the boundary of the convex mark support. Degree one and strict convexity yield a unique global saddle. Derivative convergence follows from the coefficient pressure theorem. $\square$

1.4. A quantified regular shell class. Let $G_{H, a}$ be the centered Gaussian with covariance equal to the Schur complement after particle-number extraction. A shell sequence $\mathcal{W}_\varepsilon \subset \mathbb{R}^d$ is regular when:

$$g_\varepsilon := G_{H, a}(\mathcal{W}_\varepsilon) > 0, \quad \sqrt{\mu_\varepsilon} \partial \mathcal{W}_\varepsilon \text{ has uniformly bounded Lipschitz charts,}$$

and smooth upper/lower approximations have Fourier error $r_\varepsilon = o(g_\varepsilon \mu_\varepsilon^{-d/2})$. Centered expanding boxes with $\delta_\varepsilon \downarrow 0$ and $\sqrt{\mu_\varepsilon} \delta_\varepsilon \rightarrow \infty$ are regular and have $g_\varepsilon \rightarrow 1$.

Theorem 1.5 (Mixed canonical shell coefficient). *Under the exact- N_ε , source- H saddle law,*

$$\mathbb{P}_{N,H}(\sqrt{\mu_\varepsilon}(C - a) \in \mathcal{W}_\varepsilon) = g_\varepsilon(1 + o(1)),$$

and the unscaled coefficient equals

$$\frac{e^{-\mu_\varepsilon I_{N,H}(a)}}{(2\pi\mu_\varepsilon)^{d/2} \sqrt{\det \Sigma_{N,H}}} G_{H,a}(\mathcal{W}_\varepsilon) (1 + o(1)).$$

The error is relative to the displayed Gaussian mass.

Proof. Fourier inversion uses only the continuous variable u . On the central ball, the coefficient pressure Taylor expansion gives the Gaussian. The compact annulus and high-frequency integrals are $o(g_\varepsilon \mu_\varepsilon^{-d/2})$ by Theorem 1.3 and the defining approximation error. The saddle exponential gives $I_{N,H}(a)$. No grand-canonical atom remains. $\square$

1.5. Microcanonical transfer. Let $\mathcal{S}_\varepsilon(N, a, \mathcal{W}_\varepsilon)$ be the primitive hard-core shell. Put

$$Q_\varepsilon^{\text{mc},a}(H) = \frac{1}{\mu_\varepsilon} \log \mathbb{E} \left[e^{\mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle} \mid \mathcal{S}_\varepsilon \right].$$

Theorem 1.6 (Source-dependent microcanonical pressure). *Uniformly on the complete B2 bounded-source chart,*

$$Q_\varepsilon^{\text{mc},a}(H) \longrightarrow Q_N(H, \lambda_{H,a}) - \lambda_{H,a} \cdot a - Q_N(0, \lambda_{0,a}) + \lambda_{0,a} \cdot a.$$

The initial fluctuation covariance is

$$\Sigma_{\text{mc}} = \Sigma_{\text{can}} - \Sigma_{\text{can}} C^* (C \Sigma_{\text{can}} C^*)^{-1} C \Sigma_{\text{can}}.$$

Proof. Apply Theorem 1.5 to the numerator at its own finite saddle $\lambda_{\varepsilon,H,a}$ and to the denominator at $\lambda_{\varepsilon,0,a}$. The relative shell coefficients cancel at exponential scale and remain controlled at polynomial scale. Taking the ratio gives the formula. Two differentiations of the saddle equation give the Schur complement. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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